CHAPTER 3: Setting the Boundaries of Alternative Agricultures: Defining Organic Agriculture Across Social, Agronomic and Scientific Dimensions

Abstract

Alternative ecological farming strategies are characterized by diversity and adaptation to local circumstance. These characteristics make it tricky to define and police the boundaries of these strategies. All efforts to change agriculture and food systems operate on multiple dimensions — political economy, agronomic practices and science and evaluation. These dimensions are interconnected, and each dimension's importance for delineating the boundaries of alternative farming strategies varies across movements and time. Here we outline a framework for understanding these multiple dimensions of food systems movements, argue for its importance in understanding ecologically based alternatives and apply it to the history of the organic farming concept. While initially defined as a different way of looking at farming systems, organic farming has increasingly become identified with a set of farming practices, largely defined by the set of techniques that are prohibited. This innovation has made organic farming legible to bureaucracies and facilitated its promotion through differentiated markets and public subsidies, while also eroding democratic and popular control of the concept. Many other alternative agriculture concepts have not coalesced around an algorithmic process for dividing farms who fit within that concept and those which do not. This stymies efforts to directly promote these modalities through bureaucracies but protects the aspects of these strategies built around democracy, participation, and traditional and tacit knowledge. Alternative agriculture movements face a tension between the usefulness and simplicity of rigorous and replicable categories, and their politics of ecology and participation.

3.1. Introduction

Is Climate-Smart Agriculture? This question may not be grammatical, but its answer is important to human and ecological well-being, as are the answers to many similar questions. Agroecologists M. Jahi Chapelle and Juliette Majot argue that "Climate Smart Agriculture Isn't. Agroecology is." (Chappell, & Majot, 2018)

In this odd turn of phrase, they make a subtle and crucial point about agricultural definitions and boundary-drawing: approaches to agriculture are often simultaneously defined as sets ecological and social goals, and also social and agronomic tools for meeting those goals. Climate Smart Agriculture is used to label a

governance framework, a set of farming systems, and a set of desired ecological and social outcomes. This makes it not only sensible, but crucial to ask: Does Climate-Smart Governance produce Climate-Smart Outcomes? Is Climate Smart Agriculture (in practice) Climate-Smart (evaluatively)?

Similar questions abound. Is sustainable agriculture sustainable? Is sustainable intensification? Does regenerative agriculture regenerate? Or, in the theme of this paper- "Is organic agriculture holistic"- in the parlance of the early 20th century, is it "organic"?

Wezel and colleagues (2009) argued that agroecology has multiple meanings; in different parts of the world it refers to either a movement, a science or a practice, or some combination. Rivera-Ferre (2018) disagreed, describing agroecology as unified, but echoed that every manifestation of agroecology has dimensions of "management," "practice(s)" and "assessment." Here, we build from these frameworks to assess alternative agricultural movements generally, arguing that any effort in food systems reform/revolution may emphasize three major dimensions: the social, the agronomic and the scientific. In other words, advocates may present their concepts as a distinct way of *organizing food-systems actors*, a distinct *way of farming* and/or a distinct *way of understanding and evaluating food and farming systems*. The emphasis on these dimensions may vary across time and space, but all will be present, even if only implicitly.

In this paper, we first demonstrate that boundary-confusion is a ubiquitous and crucial issue for making food systems more sustainable and equitable. Next, we describe a framework for thinking about these boundary confusion issues. Then, we

apply this framework to the history of organic agriculture in the US and the UK, demonstrating how boundaries shifted, not just in their content, but in their dimensions. We argue that organic farming has evolved from being primarily a scientific and philosophical perspective, to a political and economic critique, to a well-specified set of production systems. Finally, we draw lessons from this history for efforts to promote other concepts.

3.1.1 Why worry about definitions?

Agriculture stands at the nexus of enormous environmental and social challenges. Food systems are a key driver threatening the stability of the earth's ecological functioning, through biodiversity loss, deforestation, greenhouse gas emissions, and eutrophication (Campbell et al., 2017b). At the same time, over 1 billion people are food insecure, and diet-related diseases are a major source of morbidity. In light of these challenges, there is broad consensus that societies must change how food is produced, distributed and consumed (De Schutter, 2010; De Schutter & Vanloqueren, 2011; Foley et al., 2011; Godfray et al., 2010; Steffen et al., 2015b; Tscharntke et al., 2012).

Politicians, activists, and businesses offer a wide variety of programs to alter food systems to meet these challenges. These differ in both the dimensions of the food system targeted, and in what direction to change them. One major axis of debate is over minimizing the land footprint of agriculture ("sparing"), versus increasing ecological health on agricultural lands ("sharing") (J. Fischer et al., 2014). Another is political economy; either food and land markets are insufficiently liberalized (Anderson et al., 2005; Tokarick, 2008), or food systems dominated by markets, corporations and the

profit motive are always unjust and unsustainable (Magdoff, 2015). Others argue for the crucial role of modern productivist technologies such as agrochemicals, transgenic organisms, and mechanization. Some argue that these are dramatically overused (McIntyre, 2009); others argue they are underused (Avery, 1995; Paarlberg, 2009).

Distinguishing between approaches is important for making progress: clear categories of farming and food systems are foundational to individual and collective action to shape food systems. Conscientious consumption requires categorization to create useable heuristics for food consumers. Activists urging institutions to shift spending must carefully choose definitions and metrics. Research, extension, and outreach activities require clear metrics and definitions to design and evaluate programs. Researchers comparing different modalities need clear definitions to study the right systems and to aggregate evidence in meta-analyses. Policymakers and citizens need well-bounded conceptions so they can weigh different agricultural futures.

Not only is boundary-drawing an important problem, but it is a subtle one as well. Some efforts to improve agri-food systems are built upon explicit, well-bounded categories of food production systems and food products. These efforts implicitly strive for a "sorting algorithm" -- a putatively objective, rule-based procedure for distinguishing foods, farming systems and food systems which meet the definition from those which do not. Systems that fit the definition are eligible for subsidies, alternative marketing, or other rewards. But terminology for farming systems is rapidly growing, constantly changing, and riven with conflict. Further, many proposed alternative agriculture movements reject this strategy. Boundary ambiguity may have impeded the

promotion and adoption of otherwise promising strategies, especially those based on ecology and biodiversity. This makes understanding boundary-drawing critical.

Actors proposing reforms to food systems or alternatives to dominant agricultural systems describe these changes with an ever-growing array of terminology. New terms are used to distinguish products and techniques, for marketing, agronomic extension, subsidies, regulations, scientific research programs and/or for organizing social movements. While some terms have become clearly associated with one particular vision for future food-systems, many terms are adopted by opposing groups to mean different things. The term "sustainable agriculture" stands out in this regard. As in other domains, "sustainable" has been adopted by corporations and revolutionaries alike (Cronin, 2015), describing approaches ranging from low-input peasant agroecology to biotechnology-driven intensification. Sometimes sustainable simply seems to mean "good farming" (Thompson, 2007).

Even before the recent proliferation of terminologies, the boundaries of alternative agriculture strategies have often been unclear, especially for ecological or biodiversity-based approaches. These approaches are often grounded in nuance, context and thoughtful integration of multiple components. That these approaches require attention to subtle details ("a high ratio of eyes to acres" in words of Jackson & Berry (2009)) and are used "without a recipe" (Lyon et al., 2011) undermines outsiders' ability to easily understand whether a specific ecological agriculture strategy is being used on a specific farm. The same processes that support resilience, sustainability and ecological efficiency also can make a farming system less intelligible to outside scientists and bureaucrats (Scott, 1998; Vandermeer & Perfecto, 2017). A farming

strategy based on context, relationship and nuance can only be defined in a way that appears vague from the outside.

Farmers, policymakers, extensionists and scientists worldwide are often confused and misdirected by this lack of clarity. For example, farmers and extensionists often confuse silvopasture, a widely acknowledged best-management practice, with unmanaged forest grazing, which can produce extensive degradation (Arbuckle, 2009; Orefice et al., 2017; Orefice & Carroll, 2017)[5]. Researchers studying the System of Rice Intensification (SRI) dispute whether it is a recipe whose results replicate poorly (Sheehy et al., 2005), or a set of principles applied by smallholders in different ways across different contexts (Glover, 2011). Similar problems are seen in mainstream ecological approaches to more efficient farming — organizations promoting Integrated Pest Management (Bottrell & Schoenly, 2018) and Conservation Agriculture (Giller et al., 2009) have difficulties defining boundaries for these ideas, and thus promoting them, and measuring adoption. Terminology for adaptive rotational grazing systems is especially messy, slowing farmer adoption and muddying scientific assessments (Allen et al., 2011; Briske et al., 2014; Gerrish & Ohlenbusch, 1998; Gosnell et al., 2020; Hodgson, 1979).

Boundary-drawing is thus an important and subtle problem for any alternative food movement, and especially if the strategies they promote rely heavily on biodiversity. Without a definition legible to outsiders an approach will struggle to attract institutional support. Advocates will face difficulty measuring the success of

[5] The USDA National Agroforestry Center defines silvopasture as "the deliberate integration of trees and grazing livestock operations on the same land. These systems are intensively managed for both forest products and forage" (National Agroforestry Center, n.d.). In other words, the distinction between silvopasture and other forest grazing is that silvopasture is managed deliberately and with intention. This distinction is important, but also exemplary of the kinds of ambiguity dealt with in this paper.

promotion, and may encourage researchers to study the wrong things, or farmers to use practices that are counterproductive. However, the complex ecological nature of biodiversity-based strategies makes drawing clear boundaries difficult.

3.1.1 Principles and Delineation in Agriculture

Alternative agriculture approaches can be framed as principles-based. Approaches articulated in terms of principles include Organic Agriculture (Luttikholt, 2007), Agroecology (Wezel et al., 2020), Integrated Pest Management (Barzman et al., 2015), the System of Rice Intensification (Glover, 2011) and Conservation Agriculture (Hobbs et al., 2008). Principles provide heuristics for decision-makers to evaluate decisions and courses of action, allowing individuals to link management decisions to a vision of agriculture's proper relationship to society and the environment.

While principles are useful for assessing the condition of a farm or food system, a good set of ecological farming principles will be too general for algorithmic sorting of farms into those that are and are not examples of a system. The flexibility of principles across circumstances means that good-faith actors may disagree whether those principles are manifested in a particular instance. One source of disagreement can be the comparative nature of principles. As shown in Table 2, principles in agroecology often reference comparisons with other farming systems or with previous states of the same farming system[6]; implementation requires choosing a point of reference for making comparisons. Alternately, other principles utilize the language of optimization, which lacks a clear definition across contexts. These difficulties are compounded by

[6] For instance 3 of 5 principles of agroecology given by Altieri (2015) and 5 of 6 listed by Migliorini and Wezel (Migliorini & Wezel, 2017) use some form of the word "enhance." (Table cites: (Barrios et al., 2020; Delvaux, 2018; Silici, 2014)

potential tradeoffs between principles. For example, herbicide and fertilizer can be useful for increasing groundcover and reducing soil loss, but minimizing their use is also a principle. Furthermore, principles have implications across all levels of food systems- the successful application of principles in one dimension of the system may face tradeoffs in other dimensions.

Table 2: Principles of Agroecology Based in Optimization or Comparison.

Principle Type	Examples	Source
Optimization	• Agroecology *optimises* and closes resource loops (nutrients, biomass) by recycling existing nutrients and biomass in farming and food systems. • Agroecology *optimises* and maintains biodiversity above and below ground over time and space.	CIDSE (Delvaux, 2018) (Delvaux, 2018)
	• *Minimise* the use of external, non-renewable resources (including fossil fuels). • Provide the *most favourable* soil conditions for plant growth, particularly by managing organic matter and by enhancing soil biological activity.	IIED (Silici, 2014)
Improvement and/or Comparative	• Resilience: *Enhanced* resilience of people, communities and ecosystems is key to sustainable food and agricultural systems. • Diversity: *Diversification* is key to agroecological transitions to ensure food security and nutrition while conserving, protecting and *enhancing* natural resources.	FAO (Barrios et al 2020)
	• Agroecology *enhances* positive interaction, synergy, integration, and complementarities between the elements of agro-ecosystems and food systems	CIDSE (Delvaux, 2018) (Delvaux, 2018)
	• *Strengthen* the "immune system" of agricultural systems through enhancement of functional biodiversity—natural enemies, antagonists, etc., by creating appropriate habitats	Migliorini & Wezel (2017)
	• *Diversify* species and genetic resources in the agro-ecosystem. • *Enhance* beneficial biological interactions and synergies among the components of agrobiodiversity	IIED (Silici, 2014)
Both	• Securing *favorable* soil conditions for plant growth, particularly by managing organic matter and *enhancing* soil biotic activity • *Enhance* recycling of biomass and optimizing nutrient availability and balancing nutrient flow. • *Minimizing* losses due to flows of solar radiation, air and water by way of microclimate management, water harvesting and soil management through *increased* soil cover.	Altieri (2015)
	• *Enhance* the recycling of biomass, with a view to *optimising* organic matter decomposition and nutrient cycling over time • *Minimise* losses of energy, water, nutrients and genetic resources by *enhancing* conservation and regeneration of soil and water resources and agrobiodiversity	Migliorini & Wezel (2017)

Building off the frameworks of Wezel et al (2009) and Rivera Ferre (2018) we argue that the boundary work of defining alternative agriculture is conducted across 3 dimensions: social organization, agronomic practices and science and evaluation. Most alternative agriculture approaches define themselves based on criteria drawn on all three of these dimensions. As we demonstrate in the case of organic agriculture, the relative emphasis on these can shift over time, and in practice, one dimension is often dominant.

Science: Ways of thinking about and Evaluating Farming Systems

Moving from principles to evaluation and science of agricultural alternatives is conceptually simple. A set of agricultural principles points towards a vision of how systems ought to function. Practitioners propose new aspects of agricultural systems to measure and study. These can form the basis for new research agendas in agronomic science. Once these aspects can be measured, these measurements could be used define the boundaries of the strategy. In other words, people working to improve agricultural systems can set goals for agricultural systems, then use science and measurement to evaluate farms against those goals and apply a specific term only to farms that perform acceptably well.

Measurement of farm systems performance against benchmarks drawn from research provides an intuitive basis for a sorting algorithm to define an alternative agriculture paradigm. Terms such as “sustainable”, “regenerative” and “climate-smart” agriculture evoke this evaluative and scientific framing. When someone labels a farm “sustainable” or “regenerative,” they appear to make empirical claims or predictions

about performance on some environmental criteria. Despite this appearance, these concepts are usually non-evaluative, or used to describe general performance of practices, rather than performance of specific farms. For example, an evaluator may label a farm "sustainable" because it uses no-till practices, which often improve soil-health, regardless of whether the intended soil-health outcomes were achieved on the farm. The paucity of performance-based boundary-drawing is due to difficulty inherent in measuring ecological performance on individual farms (Duru et al., 2015), a problem found across sustainable land-management programs (Leimona et al., 2015).

Framing an alternative as a research program, or unique perspective on agricultural systems does not lend itself to a sorting algorithm. At the same time, it can unify disparate groups around a vision of what elements are important in a farming system and provide inspiration for experimentation and progress towards farming systems supporting that vision.

Socio-Political: Ways of Organizing People in the Food System

An alternative can also be defined by the politics and social processes that form the social context of agriculture. Algorithmic sorting on farm characteristics conflicts with values of participation and democracy central to many radical food movements. For instance, agroecological movements often adopt "constructivist" views of farming systems and knowledge transmission - that there are multiple valid "knowledges" of farming systems which must be brought into "dialogue" with one another (Mier y Terán Giménez Cacho et al., 2018). This perspective is hard to square with building objective, external definitions of farming systems. Some movements have created rule-based boundaries drawn using social dimensions, however.

Many frameworks focus on larger-scale social processes, as in the case of global peasant movements fighting for land reform and food and seed sovereignty. Most prominently, agroecology has come to represent, in the words of Miguel Altieri "the technological flag of the resistance movement (Kawell, 2007)" through an agenda of "food sovereignty." Food sovereignty, the right of peoples "to define their own food and agriculture systems" (La Via Campesina, 2007), centers the social and political processes of decision-making, rather than details of farm practices or ecology. Food sovereignty operates at the "constitutional" level of discourse (Ostrom, 2011), focusing on "who decides?" and "how?" rather than "what practices?". Political agroecology thus does not rely on an external, "objective" definition of agroecological practices or performance, but rather defines agroecology by the forms of collective decision-making made in food and agriculture. This perspective is obviously hostile to a bureaucratic, algorithmic view of what farms do or do not belong within a category of farming systems- this is a question to be made collectively.

There are numerous examples, however, of firmer boundary-rules for sorting farms in or out based on social criteria. "Holistic Management" (Savory & Butterfield, 1999) and Permaculture (Mollison & others, 1988), are both primarily defined by individual and collective process of system design, farmer learning and decision-making. Educational processes can play an important role in delineation- the central organizations in both these movements are educational non-profits who attempt to govern the use of these words by accrediting teachers, curricula, and consultants[7].

[7] Permaculture organizations often claim that the word "Permaculture" is proprietary, only graduates of an accredited course of study can use it to describe their teaching, design or farming business (Harano, n.d.; Pacific Coast Permaculture, 2000; The Northern School of Permaculture, n.d.). This appears false, it has never been an approved trademark within the US (United States Patent and Trademark Office, n.d.).

Other socially-based categories, such as “Family Farm,” “Fair Trade” or “Local Food” may be defined in a rule-based way by governments or non-profits, for differentiated marketing or determining eligibility for programs.

Practices: Ways of Farming

Agricultural alternatives are often defined by the practices which are used in them. Some modalities defined by practices are narrow in scope. For instance, agroforestry systems take many different forms, but a single practice- integration of trees with other agricultural elements – defines the category “agroforestry”. Other approaches are defined largely by excluding practices. This is prominent in food labelling, both narrowly, such as “cage-free eggs” or “non-GMO” foods or broadly, as in certified organic agriculture. Their use in food-labeling schemes suggests that exclusions of practices can be very well-suited to sorting algorithms.

Some practice-based delineations are attached to suites of practices that farmers choose from. Agroecology (Wezel et al., 2014), regenerative agriculture (Lal, 2020) and sustainable agriculture (O’connell, 1992) are sometimes characterized as suits of practices which farmers draw from to meet their agricultural and environmental context. Defining an approach this way creates only weak boundaries- it is rare to find a farm that does not use *any* ecologically-based strategies; crop rotation, no-till and cover-cropping are commonly found on largescale “industrial” farms. Some certification schemes, including Fair Trade and Rainforest Alliance, utilize scorecards, wherein farms must a earn certain threshold number of points from optional practices to

pass, thus operationalizing their vision of "more sustainable agriculture" through a suite of practices.

Marking the Boundaries of Biodiversity-Based Farming:

Operationalizing principles as definitions is particularly complicated for biodiversity-based farming systems. The political and social aspects of these approaches emphasize participation and adaptive management as critical to an ecological and just food system. The scientific and evaluative aspects emphasize the heterogeneity and complexity of sustainable agricultural systems (Vandermeer & Perfecto, 2017), meaning that great flexibility is needed in implementation. These elements reduce the external legibility and clarity of alternative agricultural practice. When local actors have leeway to interpret principles, weigh their relative importance and adapt them to local conditions which may not be externally visible, outside actors may struggle to categorize and assess these systems and their compliance with those principles. This is especially problematic if outsiders, including regulators or consumers, need categorization and suspect that farmers may be behaving dishonestly or cynically.

In such systems, the conflation of evaluative and agronomic practice-based boundary-drawing is common and can create confusion. For instance, no-till is commonly described as a regenerative, sustainable or climate-smart practice, but researchers are divided over under what circumstances, and to what degree, it improves soil-health indicators, especially carbon sequestration (Baker et al., 2007; Baudron et al., 2012; Giller et al., 2009; Luo et al., 2010; Powlson et al., 2014; VandenBygaart,

2016). Furthermore, no-till is associated with monoculture, herbicides and other practices with environmental impacts that may offset its benefits (Altieri, 2005). Similarly, Phalan and colleagues (2007) argue that defining "sustainable agriculture" as both a set of practices and a set of outcomes biases the sample for reviews of the effectiveness of those techniques (e.g. Pretty et. al. (2006b)); initiatives where these techniques are attempted but perform poorly may not be reported or included- it is ambiguous whether these are instances of "sustainable agriculture." These difficulties are magnified in biodiversity-based systems; the lack of recipes means that "the right practices" will not mechanically ensure desired results.

Converting the principles of alternative agriculture into agreed-upon criteria for boundary-drawing is not a simple task. Principles are put into practice along socio-political, agronomic, and outcome-based dimensions. Most modalities of alternative agriculture span these 3 dimensions, but which dimensions are most emphasized is not always obvious from casual observation. Next, we examine the history of organic agriculture, one of the oldest alternative agriculture movements, and how the dimensions on which it is defined have shifted over time.

3.2. The Case of Organic Agriculture

Like other alternative agriculture concepts, the philosophy of organic agriculture is often articulated as a set of principles. Recently, most sets of organic principles converged in their meaning. Alrøe and colleagues (2002) identified three principles: "cyclic principle,"- farming systems "should emulate and benefit from nature's systems and cycles, fit into them, and help sustain them", the "nearness principle"- food systems

should be participatory and localized and the "precautionary principle,"- "that action should be taken to prevent harm" before definitive evidence of harm, grounding the organic agriculture vision. The "Precautionary Principle" is also included by Benbrook and Kirschenmann (1997) along with the "Ecological Principle"- farming "should fit into and benefit from nature's systems"- and the "Systems Principle" - that the impact of technologies and practices should be evaluated holistically. The International Federation of Organic Movements (IFOAM) articulates the principles of organic farming as Health, Ecology, Care and Fairness (Luttikholt, 2007). These views broadly overlap with one another; differing more in how the vision is divided into individual principles than the vision itself.

Consensus about the principles of organic agriculture, however, is insufficient to divide farming systems into organic and non-organic categories. Adjudicating these principles may involve tradeoffs, or judging different thresholds. For instance, the exact spheres and thresholds for applying the precautionary principle are contested (Kriebel et al., 2001) and defining farming techniques as "natural" versus "unnatural" is a complex question (Verhoog et al., 2003). The importance of these ideas has remained relatively stable in organic agriculture, but the means by which these ideas have been converted in demarcation criteria have changed drastically.

3.2.1 The Organic Perspective: Science and Agronomy 1910-1960

In its early years, the organic movement defined itself primarily on a distinct way of looking at farming systems: a different set of processes to be studied and a different set of outcomes to measure than the status quo of agricultural science. The movement for

organic agriculture was by far strongest as a discursive and research community, or "critical community" (Reed, 2010). The early organic movement was distinguished by its perspectives about what aspects of agricultural systems are important, whose knowledge should be valued in understanding them and what agricultural research should look like. It was less unified as a distinct way of farming or managing food systems.

The philosophy of organics pointed to a new set of important processes, outcomes and indicators for farm systems. Howard (2010; 1946) and other thinkers (e.g. Albrecht, 1975) argued that agronomy over-emphasized soil chemistry, and especially the three soil macronutrients, at the expense of broader conceptions of soil fertility including biological and physical aspects. Organic advocates likewise expressed skepticism of chemical pest control, both due to harms from these techniques, and the belief that chemical cures mask the underlying pathologies which cause pest and disease problems (A. Howard, 2006, p. 146).

Early organic thinkers also expressed new ideas about the types of knowledge about farming systems which ought to "count" or be considered. These thinkers advocated for listening to the traditional knowledge of farmers who might lack formal education. Howard's work on the Indore method of composting is an early example of ecologically minded agronomists taking traditional knowledge and practices seriously. Since this time, organic advocates have argued that the complexity in agro-ecosystems requires engaging seriously with tacit knowledge of farmers, even when expressed in ways that contradict basic parts of the scientific worldview (Kilcher, 2007; Luttikholt, 2007). This orientation has become central to other alternative agriculture movements, especially agroecology (Méndez et al., 2013).

Organic philosophy generated several critiques of agricultural research as well as new directions. Howard (1946) criticized experiments at Rothamstead for their reductionist approach and for not examining earthworms, soils, livestock and other indicators of system health. Likewise, Balfour (1943) argued that side-by-side trials, like those at Rothamstead, could not provide insight into impacts on the whole-farm system, as beneficial soil-dwelling animals could travel between plots. The holistic conceptions of human, ecological and social health led to research projects on the nutritional quality of food from different cropping systems, whether through animal-feeding trials or direct measurement (McCarrison, 1926).

Advancing the science and philosophy of organics was the central purpose of early organic institutions. When the Soil Association was founded in 1943, its primary purpose was as a "clearing house" for organic farming information and to supervise organic farming research (Reed, 2010, pp. 55–57). For more than a decade the Haughley Experiment, which sought to prove benefits from organic farming, took up most of the organization's energy and finances.

While the organic farming movement in the United States was much less well-developed, it followed a similar pattern. The first institutional manifestations of organic farming in the US were focused on promoting and elaborating the idea of organic farming — the magazine "The Organic Farmer" (later to become Organic Farming and Gardening) founded in 1942 and the Soil and Health Foundation (later the Rodale Institute) which promoted organics research, both founded by J. I. Rodale. As we discuss later, this did not develop into a singular definition of organic methods until decades later.

Early organics did not constitute a distinct or unified critique of the social, economic or political dimensions of food systems. Instead, social and political perspectives were variable, united by general anti-urban and anti-modern sentiments. In the UK, founding leadership of the Soil Association in 1945 was dominated by monarchists and other right-wingers; the first editorial secretary had previously belong to the British Union of Facists (Reed, 2001). Other leaders, such as Lady Eve Balfour, had moderate, yet idiosyncratic politics, and some early members were leftists, including E.F Schumacher. This ideological eclecticism was common in early 20th century English agrarian reform movements (Conford, 2002). Early advocates in the United States were likewise motivated by many ideologies, including religious, spiritual and/or romantic (Bromfield, 1949; Leopold, 2001), leftist and pacifist (Nearing & Nearing, 1989) and/or distrust of modern chemistry and medicine (J. I. Rodale, 1945). This wide range of views made it difficult for organic agriculture to have meaningful political engagement, even in the UK, where members were well-networked to the political elite.

In terms of practices, organic agriculture was only weakly distinctive during this time. A few elements, such as mixed crop-livestock farming, crop rotation and composting were widely promoted and used, though the first two were still widespread among conventional farmers. While Howard argued for the complete exclusion of synthetic fertilizers, other organic advocates saw synthetics in a supplemental role within organic fertility plans (Heckman, 2006) or as a necessary compromise for profitable farming (Reed, 2003, p. 18). Indeed, Howard refused to participate in the Soil Association because it refused to take a hard line on this issue (ibid). Likewise, early organic advocates were conscious of the harms of excessive tillage, but did not adopt any

standards (Vogt, 2007). Tensions existed between pro-vegetarian beliefs in related Food-Reform movements of the turn of the 20th century, and the insistence that livestock were essential to sustainable farming (ibid). Similarly, while many organic advocates opposed all synthetic pesticides, consensus on this issue was not reached until the 1980s[8].

3.2.2 Organic agriculture and social change: 1960-1980

Organic agriculture entered the 1960s largely marginalized in the English-speaking world. Concerns about soil depletion had waned, use of agro-chemicals was rapidly increasing, and modern farming was being exported by Western governments in "the Green Revolution". A new set of environmental and political concerns, however, allowed organic agriculture to capture the zeitgeist and gain new adherents. In the 60s and 70s, citizens and activists became increasingly skeptical of "big" science, business, and government, and expressed this through environmental and social activism including anti-pollution, anti-nuclear and anti-war movements. Famine in low-income countries and the energy crisis re-invigorated Malthusian resource-scarcity concerns. In these years, activist in the US and UK became interested in organics and organic agriculture became identified with the counter-cultural left.

In the 1960s into the 1970s, organic agriculture became increasingly intelligible as a distinct set of critiques of the social, economic and political organization of food systems, through linkages to left-wing movements in the US and throughout Western Europe. Environmental problems, the Cold War, nuclear power and the upheaval of the 1960s contributed greatly to distrust in institutions across the West. In the realm of

[8] Reed (2003 p. 16) notes that in 1952, a Soil Association Member's Bulletin felt the need to address members who may "consider that the use of DDT is still justified."

agriculture, this distrust was manifested as skepticism of the safety, nutritional quality and sustainability of the increasingly industrialized food system. The publication of *Silent Spring* in 1962 (Carson, 2002) galvanized massive public opposition to pesticide usage at the height of mainstream agronomy's self-confidence in yield-enhancing measures (Beyl, 1992), exemplifying a broader set of concerns on the "New Agenda" (Youngberg & DeMuth, 2013). Organic agriculture was a ready alternative for this crisis of confidence.

Organic Agriculture also became associated with a wave of "back-to-the-landers"; young people who had not grown-up farming and sought simpler, more egalitarian lives, often in collective arrangements. These new farmers were motivated by ideologies mixing Jeffersonian Democracy, Anarchism, Marxism and Romanticism (Edgington, 2008). While the vast majority of those new farmers did not remain on the land, some who remained created successful agricultural businesses. The ideology was more persistent as the foundation for the boom in natural foods coops in the early 70s (ibid.). By the late 1960s, organic agriculture found itself at the nexus of counter-culture, the New Left and the environmental movement, and numerous grassroots organizations emerged to carry its banner (Haydu, 2011). Organic gardening clubs and natural food coops became hubs of community political activity, especially around environmental issues (Aschemann et al., 2007).

By the late 1960s, organic agriculture in Britain was experiencing a similar shift. Left-environmentalists had taken over the leadership of the Soil Association, including economist E.F. Schumacher as president and activist-scientist Barry Commoner as vice president. Schumacher linked organics to his ideas about organizational scale, appropriate technology and the downsides of economic growth (Schumacher, 1973) and even

advocated abolishing private farmland ownership, but this never became the official platform. Commoner linked organics to the technological precaution found in anti-nuclear activism. At the same time, a wave of back-to-the-landers with left-wing views organized new organizations: British Organic Farmers and the Organic Growers Association (Reed, 2003).

While the critiques from organic agriculture were unified against big business and "conventional agriculture," solutions pointed in different directions. While few explicitly conservative framings were adopted, writers such as Rodale framed organics in terms of agrarian populism; emphasizing the family farm (and traditional family) as the primary unit of production, and fairer markets as the primary means of distribution. Younger entrants to organic farming who came to it through the "New Left" and counterculture often attempted farming in communal relationships and rejected traditional families. Even among leftists there were many disagreements; while Schumacher called for decentralization and communal land ownership, Barry Commoner called for a democratically planned economy (Reed, 2010, pp. 85–88). These differences meant that organic farming did not coalesce around a definite policy program, despite being clearly aligned against the status quo of big business and government.

During this time, scientific inquiry became less central to organic identity. Critiques of science became more focused on corruption and corporate control. Emphasis moved away from proving the benefits of organics- the Soil Association closed the organic-conventional comparison experiment at Haughly in 1966- and moved towards research examining technical issues facing organic farmers (Reed, 2010). Organic or "systems" approaches to agricultural science also became less distinctive, as mainstream

researchers explored reducing pesticide and energy use by examining farms from a whole-systems perspective (Huffaker & Croft, 1976).

This period also saw increasing interest in policing the boundaries of organic by definitively categorizing farming systems as organic, driven by a desire to market organic food and support organic farmers. In the US, Robert Rodale argued that while, "when you are growing organic food for yourself, your personal definition is all that counts…The lack of a standard definition of 'organic' and a means to enforce that definition has held back the marketing of organic food" (R. Rodale, 1970, p. 30). Early attempts to distinguish organic products from conventional were largely based on trust, and agreement to adhere to organic principles. These included direct-marketing, contracts, and handshake agreements. Rodale's Organic Gardening and Farming Magazine began publishing an 'Organic Foods Shopping Guide' in 1969, several years before they published a definitive standard. In Britain, the "Whole Food Society" had attempted marketing organic foods from a storefront in London, starting in the late 50's, based on grower promises to follow organic principles. The Soil Association began certifying farms in 1967 using more detailed criteria (Reed, 2010).

Efforts to better define organic farming in 1970s and 1980s demonstrate the primacy of social and ideological dimensions; "organic farming" was the type of farming practiced by organic farmers, not the other way around. Organic farming organizations in Oregon, California and the northeast developed labelling schemes to market organic food beyond direct-to-consumer channels. These farmer-led certification programs were similar to modern participatory guarantee systems (Loconto & Hatanaka, 2018), which combine elements of social and agronomic delineation. These certification programs

were structured around good-faith efforts towards an organic ideal, collaboration between farmers and inspectors in problem-solving (Mutersbaugh, 2005) and building collective identity within the community of organic farmers (Schmid, 2007). The standards were developed by committees of farmers, who dealt with new problems by discussing what an organic farmer would do in a situation (Obach, 2015, pp. 53–57). The 1980 landmark USDA report on organic farming used an "empirical" definition of organic farming- organic farming was defined by the practices of self-professed organic farmers. They found it to be a "spectrum" "overlapping" with conventional agriculture at one end (USDA Study Team on Organic Farming, 1980; Youngberg & DeMuth, 2013).

3.2.3 Institutionalization of Organic: 1980-2000

During last decades of the 20th century, definitions of organic as a distinct way of farming became increasingly precise and generalizable. Organic activists and farmers worked to grow market share and fight off imposters by certification and labeling laws, culminating in national and Europe-wide regulations. This new regulatory structure solidified "organic" as describing a set of farming systems and the food produced by them. As such, the political dimension of organic has become increasingly subsumed by conflicts over organic rules, while the scientific dimension has become dominated by comparisons between well-defined organic and conventional systems.

As organic food became a niche market, organic farmers became increasingly concerned about "organic fraud", driving interest in definitive standards (Guthman, 1998; Schmid, 2007). Fraud threatened to erode consumer trust and drive organic growers out of business. While initially farmers' organizations certified their own members, the

newly active organic food and farming movements turned their energy towards lobbying for legislation regulating organic claims. Oregon passed the first law 1973. Between 1979-1990, over two dozen states passed organic food labelling laws, greatly increasing consumer access to organic food. However, the profusion of state laws and independent standards created confusion for businesses and consumers (Obach, 2015, p. 60). Federal legislation established the National Organic Program in 1990, which required the USDA to create a single standardized definition for organic food, finalized in 2000. While the NOP was being developed, certification agencies largely converged in their standards, but the final NOP rule was as strict, or more strict, than independent standards (Fetter & Caswell, 2002). Since the NOP was established, organic food sales have seen enormous growth, but the market for organic foods has become increasingly consolidated.

The UK followed a similar course, though with important differences. Private certification began in 1967, with the Soil Association. National regulation was established in 1987 as the UK Register of Organic Foods, and soon became enmeshed in European push for organic regulation. By 1991, nearly every country in the European Union passed a national regulation on organic farming; these were harmonized into a European standard in 1991 for crops, and 1999 for livestock (Schmid & Lockeretz, 2007). The harmonized rules allow for stricter national standards for livestock, but otherwise allow for little flexibility in setting national standards. In the UK private certifying agencies can maintain their own standards to a higher level than the EU, but in practices differences are small (Padel et al., 2009).

Since the passage of the NOPA in 1990, the politics of the organic movement have become increasingly focused on the "integrity" and details of organic standards.

These conflicts began even before the establishment of the NOP, with enormous grassroots backlash to language exploring the acceptability of GMOs, sewage sludge and irradiation for organic food production (Jaffee & Howard, 2010; Obach, 2015, pp. 117–119; Vos, 2000). Small-farmer and consumer groups have organized numerous campaigns to counter what they regard as weakening of organic standards, loopholes for large corporations, and misleading marketing. These controversies have included synthetic ingredients in organic processed foods, misleading claims for "organic" body-care products, outdoor and pasture access for animals and minor controversies over prohibited materials.

The growing consensus of what practices constitute organic agriculture facilitated more scientific comparisons of organic and conventional farming systems. Side-by-side systems trials were established by the Rodale Institute in 1980, (Pimentel et al., 2005a), the University of Wisconsin in 1989 (Posner et al., 1995), the University of Michigan in 1993 (Snapp et al., 2010) and the USDA in 1996 (Cavigelli et al., 2008), among others. This profusion of experiments has allowed facilitated meta-analyses on organic farming systems, including on yield (de Ponti et al., 2012; Ponisio & Ehrlich, 2016; Seufert et al., 2012b) and ecological impacts (Jawtusch et al., 2011; Kremen et al., 2012; Mondelaers et al., 2009; Tuomisto et al., 2012).

At the same time, the scientific perspective of farms as "organic wholes" became less distinctive to organics. The term "Sustainable Agriculture" was promoted by some organic advocates as a compromise term to make organic perspectives palatable to mainstream agricultural actors, and was institutionalized in USDA programs (Youngberg & DeMuth, 2013). Likewise, more radical scientific

approaches began coalescing around agroecology, as researchers working in Latin America began to apply their theories to systems in the global North (Wezel et al., 2009). Similarly, reformist elements of agricultural science took up studying farms as whole systems, including approaches such as Integrated Pest Management (Huffaker & Croft, 1976).

3.2.4 Organic in the 21st century: The Primacy of Practice

In the 21st century, organic food and farming movements have become largely structured around "organic" as defined by practices. Advocates and farmers generally describe organic as a production system. Political campaigns around organic are centered around practices and certification. Organic farming research is now defined by its object of study—organic farms, rather than a perspective on what should be studied in farming systems.

Organic movements have organized numerous campaigns around organic standards, both pushing to close loopholes, and fighting against efforts to make them less stringent. Activists also fight to protect organic operations from contamination with materials which are excluded from organic systems. Organic food movements also fight for policy and social change to increase acreage under organic agriculture, and access to organic food.

The organic movement is now organized around "organic" as a description of specific farming systems. Both IFOAM, the international organic advocacy umbrella, and the National Organic Program define organics as "a production system" (7 CFR § 205.2 Subpart A — Definitions., n.d.; *Definition of Organic Agriculture | IFOAM*, n.d.), though

IFOAM also includes an ideological and holistic view of organics. Likewise, policy statements on organic agriculture treat it as a well-delineated set of production systems e.g. (Hoodes et al., 2010). The dominance of this frame is made most evident by the inability of those fighting it to escape it. NGOs have fought against the "corporate takeover of organics" primarily by contesting the interpretation and definition of the organic rule (Vos, 2000).

Organic movements have mobilized around numerous issues in regulating organic foods. In 1997, activist pressure forced several major revisions to the NOP draft rule, including banning transgenic organisms. Since then, activists have mobilized several more times. Several major policy disputes involved animal agriculture, especially how livestock access to the outdoors and ability to graze are regulated and how those regulations are enforced. More recently, small farmers and consumers mobilized unsuccessfully against allowing hydroponically grown crops to be labelled organic. Efforts to tighten and protect organic regulations have been led by small-farmer and consumer organizations, while larger farms, food processors and retailers have fought for laxer rules (L. King & Busa, 2017). While stricter interpretations of organics in these dimensions would clearly favor many smaller farms, these campaigns may undermine or distract from a critique of how organic agriculture is governed (Haedicke, 2016, p. 162; Vos, 2000).

The environment in the E.U., and therefore the UK, has generally been less adversarial (for discussion of the trend in Sweden see Klintman & Boström (2004)), and in some ways more strict (Kosovska, 2013). The European Parliament has revised the regulations to close loopholes, such as prohibiting hydroponics from being considered

organic (Schmidt, 2019). While regulatory capture appears less extensive in Europe, a regulatory, practice-based view of organics predominates there as well. Indeed, the existence of disparate standards has itself become a source of political conflict, both with regards to the United States (Kosovska, 2013), and recently between Britain and the EU as a result of Brexit. In general, however, organic standards have seen a remarkable degree of convergence between countries with very different regulatory cultures (Klein & Winickoff, 2011).

Outside of battles over regulatory standards, the other political activities associated with organic take the "way of farming" framing as central. Organic advocacy groups spend considerable effort encouraging research and subsidy support for organic farming (Guthman, 2014, pp. 204–206). Organic participation in GM activism is primarily driven by concerns about pollen drift and contamination of organic crops by GM plants (Kurzer & Cooper, 2007).

That numerous farmers now refer to themselves as "beyond organic," referring to agroecological practices and/or alternative distribution networks shows the degree to which "organic" is thoroughly identified with its certification. Likewise, the development of "Regenerative Organic Certification" (Rodale Institute, 2019) accepts that organic means "certified organic" and posits that it is insufficient, while the details of the "Real Organic Project" certification contests the legitimacy USDA organic, but also frames "Real Organic" as a category constructed by closing a few loopholes in USDA Organic standards (*The Real Organic Project*, n.d.).

While "organic" is now deeply tied to production standards, many in the movement rejected that framing. Standardization imposes bureaucratic culture upon small

farms and traditional communities (Bacon, 2010; Getz & Shreck, 2006; Mutersbaugh, 2002). Many object to the time and financial costs of certification, which may exceed the benefits for small farmers who direct market (Constance et al., 2008), but others object to limitations placed on farmers by certification. Many members of alternative agriculture movements engage in practices which do not meet organic standards, but which they see as consistent with organic principles and consumers' reasons for buying organic foods.[9] "Movement organic" farmers who reject certification for practice-based rather than bureaucratic reasons may use non-organic livestock feed (including non-organic food waste) or use banned products in a targeted way for livestock health problems or pest problems in ecological orchards. Indeed, a substantial minority of self-identified organic farmers in 1980 used synthetic pesticides, herbicides or fertilizers, which they tried to minimize (USDA Study Team on Organic Farming, 1980, pp. 14–15).

In these cases, populist political economy and an adaptive view of organic principles comes into a stark conflict with rigid standards. While organic standards are obviously imperfect, there is a fundamental disconnect between the idea of organic as bottom-up, locally sensitive farming in pursuit of ecological goals, and organic as a production standard. Requiring practices or excluding them reduces farmer flexibility in decision-making and potentially reduces the quality of ecological outcomes on farms where decision-making is constrained by this.

Scientific inquiry also has a very marginal role in the boundary-drawing of organic agriculture. Regulatory processes, including National Organic Standards Board, use scientific assessments of safety to determine synthetic inputs that are allowed in

[9] For example, farmer Joel Salatin, for a time one of the organic movement's most prominent authors and speakers, pioneered and promotes a "beyond organic" farming system which includes practices which render it *ineligible* for organic certification.

organic production, as well as “natural” inputs that are excluded (National List of Allowed and Prohibited Substances, n.d.). But the vast majority of research on organic agriculture now takes for granted “organic” as a well-defined set of production systems. This research compares the relative advantages and disadvantages of organic vs conventional systems and sometimes takes a whole-systems approach. The emphasis in this research and other agronomic research on soil health, energy efficiency and biodiversity represents a partial victory for an “organic” perspective on farming systems research, in the original meaning of the word.

In summary, most activities around organic farming now take for granted that it is first and foremost a distinct and well-defined way of farming (Reganold & Wachter, 2016). The political activities of the organic movement are largely centered around the details of its definition and promoting its uptake. The scientific activities related to organic farming likewise focus on developing techniques suited to organic production systems and the assessment of these production systems in comparison to conventional ones. This stands in distinct contrast to other popular alternative agriculture movements.

3.3 Discussion

3.3.1 Benefits and Harms of Current Conceptions:

The firm, practice-based boundaries around organic agriculture have had several important effects. Organic agriculture is now defined in a “bureaucratically legible” way (Scott, 1998). This has facilitated its growth but undermined its original philosophical meaning. These definitions have enabled a wide array of policy interventions and have

simplified some aspects of organic advocacy. At the same time, activists and small farmers lost control of the rule-making process, especially in the United States.

Clear boundaries for organic have enabled a very clear set of market and policy interventions. Organic agriculture is certified for premium prices and can receive specific subsidies and technical support in most countries. Public and non-profit institutions around the world have adopted measurable targets for increasing the procurement of organic food (Borsatto et al., 2020; *Signatory Schools*, n.d.), including a nationwide program in Denmark (Sørensen et al., 2015). Additionally, consensus in defining organic agriculture has allowed researchers to aggregate studies comparing agricultural systems across an enormous range of contexts (Seufert et al., 2012b). And not insignificantly, the definition is clear enough that statistics of adoption and diffusion can be kept on local, national and international scales (e.g. Figures 1 &2).

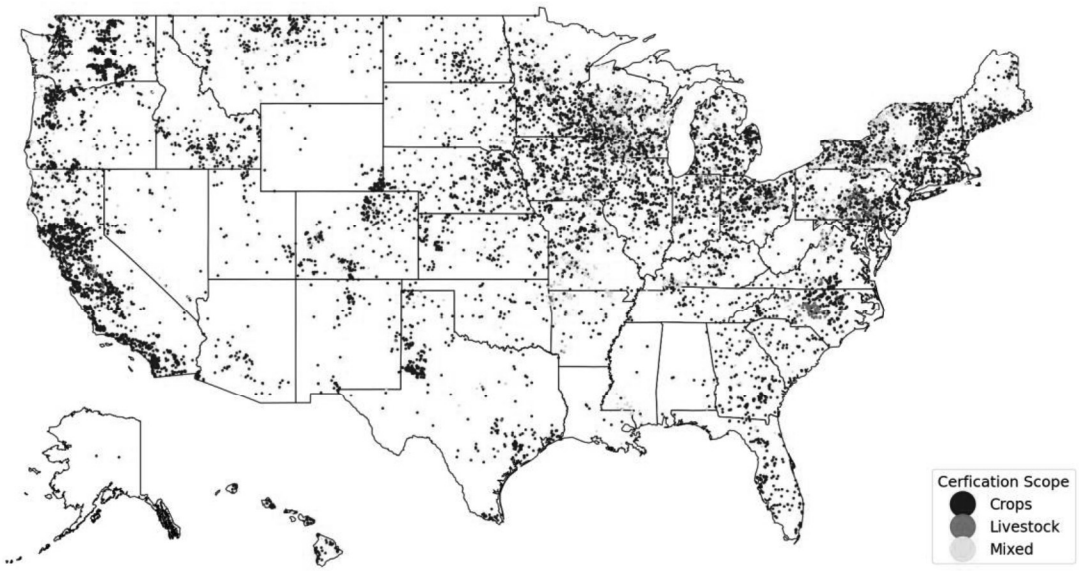


Figure 1: Every organic farm in the US as of September 2021: Data from the USDA Organic Integrity Database.

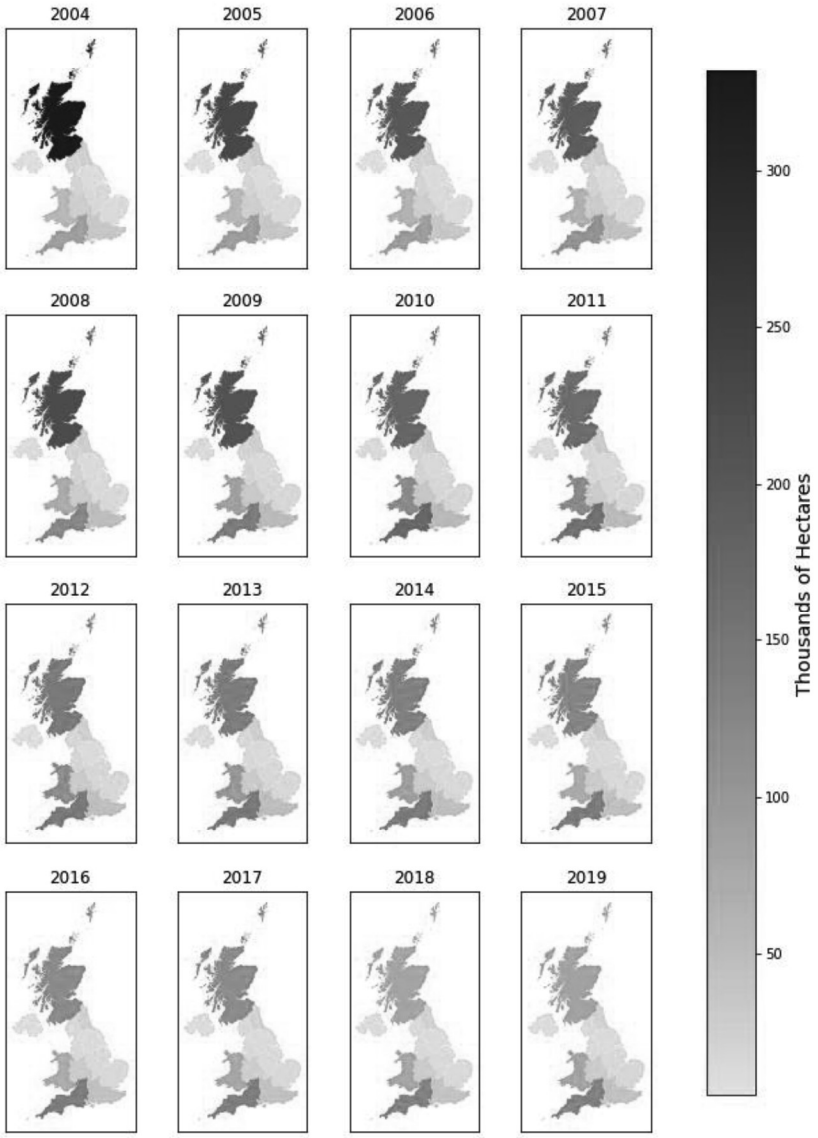


Figure 2: Certified organic Land (Thousands of Hectares) in UK administrative regions from 2004 to 2019.

In the US, contrary to popular narratives, the standards of organic production have generally become more, rather than less, technically stringent since the introduction of the NOP. This is especially true for organic livestock production, where "access to pasture" for ruminants was not required by all pre-NOP certifications, and was not rigorously defined until 2010 (Fetter & Caswell, 2002) and some advocates argue the current rule is *too strict* for some agroecological contexts (Goodman et al., 2012, pp. 168–172). Similarly, early standards for poultry were inconsistent on "outdoor access" while this became a requirement under the NOP (ibid) and certified organic livestock feed was long not required for organic animals.

What has weakened, however, is the ability of organic activists and small farmers to adapt technical standards to uphold organic principles. Activists argue that corporate actors are attempting to change the meaning of organic, to drive it away from its principles, but most "corporate capture" has come from standards not changing quickly enough to adapt to new technologies. The philosophy and movement of organic farming has long been interested in the continuous improvement of farming systems, but the National Organic Program has generally been reticent to tighten organic standards in reaction to new developments, such as new production systems that violate the spirit of organics or progress in organic techniques that make traditionally-used materials un-necessary (Obach, 2015, pp. 144–149). Organic advocates during the formation of the NOP did not conceive of a need to disallow "organic hydroponics" (Morath, 2018) and confinement-based organic livestock production. Maintaining the philosophical meaning of organics may require constant *changes* in the content of organic standards.

3.3.2 Should Movements Build firm Boundaries for their Strategies?:

Many of the most popular approaches to reforming or transforming agriculture lack the firm boundaries that have been built around organic farming. While concepts like Climate Smart Agriculture (CSA) and Agroecology may have a unifying normative vision or *gestalt*, no consensus processes exist for sorting farms in or out of these categories, and many advocates reject the possibility of such. Newton and colleagues (2020) demonstrate that practitioners and researchers of "regenerative agriculture" ground it in practices, outcomes or both. In terms of this paper, it may be a distinct way of farming, or distinct way of evaluating farming systems, though they ignore that some also include strong political-economy elements in regenerative agriculture as well (Rodale Institute, 2019; *Why Regenerative Agriculture?*, 2019). On the more institutional side of the spectrum, "climate-smart agriculture" is presented as a set of practices, a framework for governance and corporate social responsibility, and/or a set of outcomes (Newell & Taylor, 2018; M. Taylor, 2017). Critics argue this shows CSA lacks definitive content and serves as a rhetorical shield for business-as-usual (Via Campesina, 2015). In both these cases, rhetorical breadth and lack of firm boundaries is advantageous for promoting broad acceptability among a wide range of stakeholders and practitioners but does not provide bureaucracies with clear guidance in transforming agricultural systems.

Agroecology is a particularly rich example; it appears as a distinct vision of agriculture which combines a way of looking at farming systems, a way of organizing food systems and a set of farming practices in different ways across contexts, with no dimension clearly dominating. In some contexts, many are clear that inputs excluded from organic production, especially transgenic seeds, are a hard line separating

agroecology from other approaches (Pimbert, 2015). But in others, participatory approaches to farmer learning for sustainability can be considered "agroecology" even if the farm systems themselves do not earn the label (Heleba et al., 2016; Morris et al., 2013). Agroecological research and practices are concentrated in non-food export commodities such as coffee (Mason et al., 2021) — where even alternative value-chains fall short of agroecological food-systems transformation. Agroecology's boundaries are policed by several factors weighed contextually, based on a normative vision, rather than a consistently applied set of rules.

Indeed, eschewing a formalized, objective definition may be the primary distinction today between agroecology and organic agriculture. The principles which motivate the two are very overlapping, as Table 3 shows. As discussed here, grassroots politics, traditional ecological knowledge, and an ecological perspective have long been essential to organic identities and movements even if they are no longer central to policing its boundaries. Those seeking to expand institutional or governmental support for agroecology through an objective, bureaucratic definition built from its principles ought to reckon with the limitations of the current conception of organic agriculture produced from an attempt to do precisely this.

Given the large potential downsides of firm, algorithmic definitions, it is important to explore options for promoting novel alternatives that do not require them. Land redistribution and anti-monopoly reforms may create a more conducive environment for ecological farming without directly targeting farming practices. Pigouvian reforms taxing fertilizers, pesticides and fuel can promote farming systems which utilize ecological processes to replace these inputs. Lastly, governments and NGOs

can invest in the capacity of horizontal farmer-to-farmer knowledge-sharing networks (Carolan, 2006; P. M. Rosset et al., 2011), which can innovate and progress towards a more ecological agriculture without requiring a firm procedural definition of what that alternative is.

3.3.3 A Framework for Analyzing Alternative Agriculture Certification and Categorization:

There are many apparent problems and contradictions in defining firm boundaries around alternative agriculture strategies, but categories are also essential. Here we outline a framework for analyzing and improving such boundary-drawing efforts.

What are the goals of the categorization?

The aims of categorization can be divided into several categories. First, categories of farming systems are necessary for farming systems research. This holds equally true for observational research of real commercial farms, field-trials at research stations and computer-modelling studies. Categorization is important for policy: if a government wants to mandate, ban or incentivize certain kinds of farming systems, it needs a categorization rule to specify which systems are covered. Similarly, categorization plays an important economic role, including certification and other labeling schemes. Lastly, categorization plays a crucial role in growth and promotion efforts for a farming system, and their evaluation. These different goals can imply different ways of drawing categorizations. While very firm, farm-based categories are

obviously important for heavy-handed regulation, or product certification, broader, continuum-based approaches may be more important for other goals. For instance, a government seeking to promote agroecology might focus on landscape-level metrics of agrobiodiversity, soil cover, habitat connectivity and/or environmental performance rather than trying to define which farms are or are not 'agroecological.'

Who has power?

Questions of power are central to drawing categorizations of farming systems from two dimensions. Who has the power to participate in forming and maintaining a definition can have important consequences. In the USDA National Organic Program, large food processors and agribusiness interests have come to dominate the procedures of updating regulations. These interests have favored productivity and consumer concerns about pesticides and GMOs above other concerns historically important to the organic movement.

Likewise, the process of applying a definition can also create important power dynamics. Bureaucratic certification processes can be damaging to the norms of more traditional communities, but comparatively easy for large wealthy landowners or corporations. Further, the burdens of verification also represent important power differentials. In the case of certifying alternative foods, a large proportion of the price premium may be consumed by record-keeping and reporting costs. This undermines the effectiveness of these programs. In some cases, creating greater compliance and reporting burdens on environmentally harmful practices and inputs may be preferable, if there is sufficient governance capacity.

Other Questions for Understanding the Scope of a Alternative Agriculture Strategy

In addition to these bigger picture questions, several other questions can help to understand the scope of an alternative farming systems strategy, and how the term is actually being used. These questions can help to clarify whether the way that a definition is structure is logical, whether it conforms to both popular perceptions and to the goals that it is structured around.

Is this term actually used to classify farming systems and farms?

Is the term being used evaluatively?
Example: "Does sustainable agriculture mean a set of practices or a set of outcomes?"

Is this evaluative use specific or general?
Example: "Does sustainable just mean 'good farming' or does it mean a specific set of outcomes?

What sub-goals are implicit in in the components of evaluation? What ones might be left out?

Is the term used to evaluate the expected outcomes of practices or farming systems themselves?
"Are all farms that use no-till (or some other practice) sustainable agriculture by definition?"

Does the term describe a set of proscribed or prescribed practices?
If the set of practices is a suite of options, how common is it that a farm uses none of these practices?

If most farms are on a spectrum of adopting some but not all practices, how is this spectrum converted into a classification?

If the set of practices involves firm requirements or bans, are these requirements always appropriate? Do the advocates present them as such?
Example: "Is it always never appropriate to use the synthetic materials banned from organic agriculture?" "Do organic advocates agree?"

What are the explicit or implicit political, social, and economic dimensions of the term?

Are some farms included or excluded in the term based on their social or economic characteristics? What are these characteristics?

Who participates in constructing and reproducing this definition?

What political values and governance structures are promoted and enforced by the processes used to distinguish between farms that are and are not described by a term?

Table 3: Comparing the Principles of Organic Agriculture and Agroecology.

FAO (Barrios et al 2020) Agro-ecology Element	FAO (Barrios et al 2020) Text	Organic Principle	IFOAM Text
DIVERSITY	Diversification is key to agroecological transitions to ensure food security and nutrition while conserving, protecting and enhancing natural resources	Ecology	"Organic agriculture should attain ecological balance through the design of farming systems, establishment of habitats and maintenance of genetic and agricultural diversity."
CO-CREATION AND SHARING OF KNOWLEDGE	Agricultural innovations respond better to local challenges when they are co-created through participatory processes.	Care	"scientific knowledge alone is not sufficient. Practical experience, accumulated wisdom and traditional and indigenous knowledge offer valid solutions, tested by time."
SYNERGIES	Building synergies enhances key functions across food systems, supporting production and multiple ecosystem services.	Ecology	"Organic farming, pastoral and wild harvest systems should fit the cycles and ecological balances in nature."
EFFICIENCY	Innovative agroecological practices produce more using less external resources.	Ecology	"Inputs should be reduced by reuse, recycling and efficient management of materials and energy in order to maintain and improve environmental quality and conserve resources"
RECYCLING	More recycling means agricultural production with lower economic and environmental costs	Ecology	"Inputs should be reduced by reuse, recycling and..."
RESILIENCE	Enhanced resilience of people, communities and ecosystems is key to sustainable food and agricultural systems.	Health	"Immunity, resilience and regeneration are key characteristics of health."
HUMAN AND SOCIAL VALUES	Protecting and improving rural livelihoods, equity and social well-being is essential for sustainable food and agricultural systems.	Fairness	"Organic agriculture should provide everyone involved with a good quality of life, and contribute to food sovereignty and reduction of poverty"
CULTURE AND FOOD TRADITIONS	By supporting healthy, diversified and culturally appropriate diets, agroecology contributes to food security and nutrition while maintaining the health of ecosystems.	N/A	N/A
RESPONSIBLE GOVERNANCE	Sustainable food and agriculture requires responsible and effective governance mechanisms at different scales – from local to national to global.	Fairness	"Fairness requires systems of production, distribution and trade that are open and equitable and account for real environmental and social costs."
		Care	"Decisions should reflect the values and needs of all who might be affected, through transparent and participatory processes."
CIRCULAR AND SOLIDARITY ECONOMY	Circular and solidarity economies that reconnect producers and consumers provide innovative solutions for living within our planetary boundaries while ensuring the social foundation for inclusive and sustainable development.	Fairness	"those involved in organic agriculture should conduct human relationships in a manner that ensures fairness at all levels and to all parties – farmers, workers, processors, distributors, traders and consumers"

3.4 Conclusion

For decades, farmers, activists and scientists have worked towards alternative paradigms of food and farming that emphasize biodiversity and social participation. The very nature of the alternatives put forward poses large challenges for delineating their boundaries. Looser boundaries allow for flexibility, but can hamper dissemination, promotion, and evaluation.

These movements have used socio-political processes, agronomic processes, and farm-system outcomes to delineate their new, better alternatives. These dimensions are used to a greater or lesser extent at different times, but often one dimension becomes dominant. Organic agriculture passed through stages where each of these dimensions was dominant. In both the US and UK emphasis shifted from the philosophy and evaluation of farming systems, to political economy critiques of the food system into a set of farming systems. In the last several decades, “organic” has become deeply tied to the idea of an objective, bureaucratic definition of farming practices; a sorting algorithm that can separate the “organic” from the “not organic.”

Eventually the desire to rigorously divide all food and farms -- into those which are organic and those which are not --- led to an almost exclusive emphasis on agronomic practices. This has facilitated market growth and government supports for organic farming, as well as research on this topic but, to many, this has also undermined its integrity and revolutionary potential. Other movements face a tradeoff between the transparency and clarity of how they define farming systems, and the participation and flexibility embedded in their ideals.